CS341 · COMPUTER GRAPHICS

Question Bank Practice

Practice on every midterm chapter

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01 Questions

A TRUE / FALSE · 1

1. In a random scan system, the picture is stored as a set of line-drawing commands in the display list. ( )
2. Random scan systems can easily display realistic shaded scenes. ( )
3. A pen plotter is a random scan device. ( )
4. A raster scan device scans the screen from top to bottom, one row at a time. ( )
5. Printers are raster scan devices. ( )
6. In an interlaced scan, the even lines are scanned first and then the odd lines. ( )
7. A monochrome 640 × 480 frame buffer needs 307,200 bytes. ( )
8. Lower-persistence phosphors require higher refresh rates to avoid flicker. ( )
9. High-persistence phosphors are useful for animation. ( )
10. An LCD is an emissive display. ( )

A TRUE / FALSE · 2

11. In a passive-matrix display, each pixel is controlled by its own transistor. ( )
12. For crisp images the dot pitch should be less than 0.31 mm. ( )
13. A point plus a vector is a point. ( )
14. 0.3P + 0.7R is a valid point. ( )
15. The cross product is defined for 2D vectors. ( )
16. Every affine combination is a convex combination. ( )
17. RGB is a subtractive color model. ( )
18. In the CMY model the origin (0, 0, 0) is white. ( )
19. Black-and-white TV uses only the Y signal of YIQ. ( )
20. DDA uses only integer arithmetic. ( )

B MCQ · 1

1. Another name for a random scan display is:
   a) Raster b) Vector (calligraphic) c) Bit map d) Interlaced
2. In random scan systems, the picture definition is stored in the:
   a) Frame buffer b) Display list c) Bit map d) Video controller
3. Persistence is the time for emitted light to decay to:
   a) Half b) One-quarter c) One-tenth d) Zero
4. Which is NOT a hard-copy device?
   a) Film recorder b) Pen plotter c) Joystick d) Electrostatic printer
5. Screen clarity depends on:
   a) Resolution, dot pitch, refresh rate b) CPU, RAM, disk c) Color, size, weight d) Voltage only
6. The most important example of a nonemissive flat-panel display is:
   a) Plasma b) LED c) LCD d) Thin-film electroluminescent
7. Which flat-panel arrangement is brighter and sharper?
   a) Active-matrix b) Passive-matrix c) Both equal d) Neither

B MCQ · 2

8. If a = (3, 4), then |a| equals:
   a) 7 b) 5 c) 25 d) 12
9. The counterclockwise perpendicular of a = (2, 5) is:
   a) (5, −2) b) (−5, 2) c) (−2, −5) d) (2, −5)
10. In homogeneous form, a vector has a fourth coordinate equal to:
   a) 0 b) 1 c) −1 d) 2
11. Cyan ink absorbs:
   a) Green b) Blue c) Red d) Yellow
12. Which models are user-oriented?
   a) RGB and CMY b) YIQ and RGB c) HSV and HLS d) CIE and CMY
13. In HSV, pure hues have:
   a) S = 0, V = 1 b) S = 1, V = 1 c) S = 1, V = 0 d) S = 0.5, V = 0.5
14. Bresenham's algorithm is preferred over DDA because it:
   a) Uses floating point b) Uses only integers c) Draws thicker lines d) Needs no starting point
15. Which OpenGL mode draws a closed polyline?
   a) GL_LINES b) GL_LINE_STRIP c) GL_LINE_LOOP d) GL_POINTS

C FILL IN THE BLANKS

1. Each row of a raster screen is called a _____.
2. A screen point is called a _____ (picture element).
3. The frequency at which a picture is redrawn is the _____.
4. The space between the centers of adjacent pixels is the _____.
5. Two colors that combine to produce white are _____ colors.
6. Adding black pigment to a pure color produces a _____.
7. Red + Blue = _____ Cyan + Yellow = _____
8. CMYK adds K, which stands for _____.
9. The HSV color space has the shape of a _____, and HLS a _____.
10. The tongue-shaped CIE curve is called the CIE _____ diagram.
11. The OpenGL utility library has the prefix _____ and the toolkit library _____.
12. The initial Bresenham line decision parameter is p0 = _____.

D SHORT ANSWER

1. Compare random scan and raster scan (storage, resolution, color).
2. Give two advantages and two disadvantages of random scan systems.
3. Name the parts of a CRT and the job of the deflection plates.
4. What is the difference between beam-penetration and shadow-mask color CRTs?
5. Compare active-matrix and passive-matrix displays.
6. What is the difference between additive and subtractive color mixing?
7. What are the three uses of the CIE chromaticity diagram?
8. Why is black (K) added to CMY in printing?